

P1 Buy List Summary - Tunisia

What to buy for the first complete printed prototype, with direct product links and a clear budget total.

1. Exact online products

Buy	Exact product	Shop / online link	Unit	Line total
4	608-2RS bearing, 8 x 22 x 7 mm	Tunisie Roulements	3.500	14.000 TND
1	Arduino Pro Micro, ATmega32U4, 5 V / 16 MHz	CoThings product	28.000	28.000 TND
2	SS49E linear Hall sensor	Tuni Smart product	2.500	5.000 TND
1	JST 2.54 mm, 3-pin female cable + male header	Didactico product	1.500	1.500 TND
1	USB-A to Micro-USB data cable, 1 m	Didactico product	6.000	6.000 TND
1	DRS-6 heat-shrink, 6 mm, 2:1	Didactico product	2.200	2.200 TND
2	Neodymium magnet, rectangular 20 x 10 x 2 mm	Celectronix product	1.499	2.998 TND

Confirmed-price online subtotal: 59.698 TND (before delivery).

The magnet is a locally available candidate, not a drop-in fit. The released holder was made around a 10 x 5 x 3 mm block. If the 20 x 10 x 2 mm magnet is purchased, measure it with calipers and print only a revised magnet coupon/holder before the structural set.

Optional printing material

Choice	Product / planning basis	Price	What it means
If Amine has enough PETG	Use his stock after slicer confirmation	0 TND purchase	Ask for remaining grams and actual slicer estimate
Minimum new spool	LUME PETG 1.75 mm, 1 kg	86.000 TND	Covers the 685 g final-parts estimate, but little reprint reserve
Recommended development stock	Two 1 kg PETG spools	172.000 TND	Covers coupons, all final parts and useful reprint reserve

Current engineering estimates: coupons 233 g; final 19 parts 685 g; all 27 pieces 918 g. Amine's real slicer result replaces these planning values.

2. Buy locally from a quincaillerie / metal shop

Group	Exact requirement	Budget
Steel	1 x 25 x 5 x 280 mm flat bar; 2 x 25 x 3 x 180 mm reinforcement strips	13-30 TND
M8 pivot stack	1 x M8 x 90 class 8.8 partially threaded bolt; 1 nyloc; washers/shims; 8 ID x about 10 OD x 13.6 mm spacer	10-25 TND
M6 bolts	4 x 35; 2 x 40; 4 x 25; 10 x 30; 1 x 45; plus 2 rig bolts to suit rig	20-35 TND incl. nuts/washers
M6 nuts/washers	At least 25 nyloc nuts and 40 washers, including large load-spreading washers	Included above
M4 hardware	4 x M4x35; 2 x M4x30; 4 x M4x20; 1 x M4x40; matching washers/nylocs	6-12 TND
M3 hardware	5 x M3x20; 2 x M3x12; 4 x M3x10; nuts/washers; 4 x M3 heat-set inserts	4-8 TND
Spring trials	Extension springs: 15-20 mm OD, 60-100 mm hook-to-hook; buy light and medium samples	10-25 TND
Desk clamp	M10 x 100 screw/rod; M10 coupling nut about 30 mm; swivel foot; 46 mm lower rubber; 90 x 62 x 2-3 mm upper rubber	16-40 TND

Local mechanical estimate: 79-175 TND. These are planning allowances, not confirmed shop quotations.

Totals

Scenario	Estimated cash total
Hardware + electronics, Amine supplies filament	139-235 TND
Hardware + electronics + one new 1 kg PETG spool	225-321 TND
Hardware + electronics + recommended two PETG spools	311-407 TND

The total excludes delivery, tools, failed prints and optional spare fastener packs. Call/check stock before paying because web prices and availability can change.

Suggested shops

Bearings: Tunisie Roulements, 6 Rue Chedly Kallala, Lafayette, Tunis 1002. **Bolts:** SEGEMO, 10 Rue Aid Jaber, Tunis 1000. **Electronics:** CoThings, Route X near Stade Bardo; Tuni Smart, 1 Rue Pierre Mendes France, Ariana; Didactico, Cite des Martyrs, Sfax. **Magnet:** Celectronix, Centre Said, Avenue Habib Bourguiba, Megrine.

3. The eight test pieces: directory and reason

Send this folder to the printer:

C:\Users\moam\OneDrive\Documents\Handbrake-Elkateb\prototype_p1\printer_handoff_amine\01_CALIBRATION_FIRST\

Why eight? They are eight different tests. They are not eight versions of the same part. Print each file once for the first calibration run.

File	Why it is printed
COUPON_608_POCKETS_DEPTH_7.stl	Chooses the correct single-bearing pocket fit for Amine's printer/material.
COUPON_608_POCKETS_DEPTH_14.stl	Checks the deeper two-bearing hub pocket; deep holes can print differently.
COUPON_M8_CLEARANCE.stl	Selects the M8 pivot clearance using the actual smooth bolt shank.
COUPON_M6_CLEARANCE.stl	Selects clearance for the structural M6 bolts.
COUPON_M6_NUT_TRAPS.stl	Selects the nut-trap across-flats size so nuts fit without spinning.
COUPON_MAGNET_POCKETS.stl	Checks magnet retention. Revise this coupon only if using the 20 x 10 x 2 mm local magnet.
COUPON_HALL_GAP_JIG.stl	Tests Hall response at 3, 5, 8, 10, 12 and 15 mm gaps.
COUPON_LOADED_BEARING_WALL.stl	Checks representative PETG bearing-wall cracking/deformation before the full hub is printed.

After the first run

1. Test every coupon with the actual purchased hardware. 2. Record the winning hole/pocket/nut-trap sizes and send photos/measurements. 3. Only failed or changed coupons are reprinted. 4. Regenerate P1.1. 5. Print the 19 final functional parts - not the old uncalibrated set.

The number 18 refers to the later mixed-material P1-C design. The mostly printed P1/P1.1 prototype has 19 final printed parts.